Expert Systems With Applications 246 (2024) 123261

Contents lists available at ScienceDirect

Expert Systems With Applications

journal homepage: www.elsevier.com/locate/eswa

Developing conversational Virtual Humans for social emotion elicitation based on large language models

Jose Llanes-Jurado, Lucía Gómez-Zaragozá, Maria Eleonora Minissi, Mariano Alcañiz, Javier Marín-Morales *

Instituto Universitario de Investigación en Tecnología Centrada en el Ser Humano, Universitat Politècnica de València. Ciudad Politécnica de la Innovación., Camino de Vera, s/n, Edif. 8B, 46022, València, Spain

ARTICLE INFO

Keywords:
Virtual Human
Conversational agent
Affective computing
Emotion recognition
Large language models
Statistical learning

ABSTRACT

Emotions play a critical role in numerous processes, including, but not limited to, social interactions. Consequently, the ability to evoke and recognize emotions is a challenging task with widespread implications, notably in the field of mental health assessment systems. However, up until now, emotional elicitation methods have not utilized simulated open social conversations. Our study introduces a comprehensive Virtual Human (VH), equipped with a realistic avatar and conversational abilities based on a Large Language Model. This architecture integrates psychological constructs – such as personality, mood, and attitudes – with emotional facial expressions, lip synchronization, and voice synthesis. All these features are embedded into a modular, cognitively-inspired framework, specifically designed for voice-based semi-guided emotional conversations in real time. The validation process involved an experiment with 64 participants interacting with six distinct VHs, each designed to provoke a different basic emotion. The system took an average of 4.44 s to generate the VH's response. Participants assessed the naturalness and realism of the conversation, scoring averages of 4.61 and 4.44 out of 7, respectively. The VHs successfully generated the intended emotional valence in the users, while arousal was not evoked, though it could be recognized in the VHs. Our findings underscore the feasibility of employing VHs within affective computing to elicit emotions in socially and ecologically valid contexts. This development holds significant potential for application in sectors such as health, education, and marketing, among others.

1. Introduction

Affective Computing (AfC) explores the capacity for eliciting, recognizing, comprehending, and appropriately responding to human emotions. This interdisciplinary field merges insights from psychology, computer science, and biomedical engineering (Picard, 2003). Emotions play a critical role in a variety of processes, including decision-making, creativity, and social interaction. Many works have focused on discerning an individual's emotional state through biomarkers, self-report questionnaires, and machine learning algorithms. Given that AfC probes directly into human behavior, its applications span across diverse sectors, such as marketing and education, with a special emphasis on healthcare. For instance, numerous studies have investigated depression recognition utilizing a variety of biosignals (Safayari & Bolhasani, 2021) and recently, there are different works that have used voice as a biomarker of the depression level of the subject (Balano, Huerto, Sanchez, Saharkhiz, & De Goma, 2019; Suparatpinyo & Soonthornphisaj, 2023). Notably, the data collection environment for these systems is of critical importance, as it must mimic real-life scenarios to trigger specific phenomena associated with depressive symptoms, which are linked with heightened negative emotional states (Gross & Jazaieri, 2014). Other research has focused on stress recognition by analyzing physiological responses, with an aim to enhance treatment strategies and potentially prevent these conditions (Greene, Thapliyal, & Caban-Holt, 2016). Hence, the ability to evoke, comprehend, and recognize emotions poses a significant challenge, one that has far-reaching implications across numerous fields, particularly in enhancing human wellbeing.

Emotions can be quantitatively analyzed using Russell's Circumplex Model (Russell, 1980). This model postulates that every emotion is

* Correspondence to: Instituto Universitario de Investigación en Tecnología Centrada en el Ser Humano, Universitat Politècnica de València. Ciudad Politécnica de la Innovación, Access N – Building 8B – 3rd Floor Camino de Vera s/n, 46022, Valencia, Spain.

E-mail addresses: jllajur@upvnet.upv.es (J. Llanes-Jurado), lugoza@htech.upv.es (L. Gómez-Zaragozá), meminiss@htech.upv.es (M.E. Minissi), malcaniz@htech.upv.es (M. Alcañiz), jamarmo@htech.upv.es (J. Marín-Morales).

https://doi.org/10.1016/j.eswa.2024.123261
Received 15 June 2023; Received in revised form 20 December 2023; Accepted 15 January 2024
Available online 18 January 2024
0957-4174/© 2024 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

a linear combination of two affective dimensions: arousal and valence. The arousal dimension delineates the individual psychophysiological activation related to the emotion, while the valence dimension quantifies the subjectively experienced positivity or negativity of the emotion (Nardelli, Valenza, Greco, Lanata, & Scilingo, 2015). This bifurcation results in four distinct regions of the model: high arousal and positive valence correlate with emotions such as happiness or excitement; high arousal and negative valence is indicative of angry; low arousal and negative valence is associated with sadness or depression; and finally, positive valence with low arousal is characteristic of a relaxed or contented state. Emotions induce measurable physiological and behavioral changes (Ludlow, 2015). Research in voice patterns has risen in the past year in the subfield known as Speech Emotion Recognition, demonstrating a high capability for recognizing emotions (Abbaschian, Sierra-Sosa, & Elmaghraby, 2021), founding this conclusion in many different languages. Moreover, other biosignals has been used to model emotions such as eye-tracking, electrodermal activity (EDA), electrocardiogram (ECG), and electroencephalogram (EEG).

The majority of AfC research primarily centers on stimuli derived from image or audio sources. For instance, the International Affective Picture System (IAPS) dataset comprises a set of images depicting individuals, objects, or events, all of which have been standardized based on valence and arousal parameters (Bradley & Lang, 2007). Another example is the DEAP dataset, which encompasses a collection of 120 videos evaluated in terms of arousal and valence (Koelstra et al., 2012). However, the pursuit of AfC research necessitates meticulous approximation to specific, realistic daily scenarios, thus underscoring the need to develop innovative ecological experimental tools. In the landscape of such advancements, Virtual Reality (VR) has emerged as a significant and promising tool. Currently, VR heralds a paradigm shift for behavioral research in psychological assessment. It allows a reproducible experience between different users with a high degree of presence, the sensation of 'being there'. Technological strides have facilitated the manifestation of VR across diverse platforms. One application is via powerwall screens or Cave Automatic Virtual Environment (CAVE) technologies. Although classified as semi-immersive VR, these platforms nonetheless deliver a high degree of immersion without isolating the physical body of the subjects (Borrego, Latorre, Llorens, Alcañiz, & Noé, 2016; Clemente, Rodríguez, Rey, & Alcañiz, 2014). In the realm of immersive technology, Head-Mounted Displays (HMDs) offer a distinct experience. These devices provide an unparalleled degree of immersion, isolating the user from the external world and simulating a complete virtual experience (Borrego, Latorre, Alcañiz Raya, & Llorens, 2018). Owing to these characteristics, VR has emerged as a highly engaging and realistic medium for emotion elicitation. Numerous experiments employing this tool have successfully achieved their objectives in emotion elicitation. For instance, the study by Marín-Morales, Higuera Trujillo, Greco, Guixeres, Llinares, Scilingo, Alcañiz Raya, and Valenza (2018) explored emotion elicitation through four different virtual environments, effectively validating their initial hypothesis by inducing varied emotional responses in alignment with each virtual environment. Similarly, the work of Susindar, Sadeghi, Huntington, Singer, and Ferris (2019) demonstrated that VR could serve as a more efficient medium to elicit emotions such as anger or fear compared to desktop or non-immersive tools.

Another important attribute is that VR also allows the introduction of elements which are not possible to achieve in real world experimentation such as Virtual Humans (VH). VH are human-like characters which commonly interact through a computer screen and/or speakers. They exhibit human-like behaviors such as speech or gestures, but also other human characteristics as emotions, empathy or memory (Burden & Savin-Baden, 2019). For the moment, a VH is a computer program which tries to simulate a human. There is a set of main elements that build a body and the mind of the VH. On the one hand, the body aims to provide real-time audiovisual content for the VH, enabling real-time interaction, as well as the senses to receive information from the users. On the other hand, the mind aims to provide the ability to interpret natural language, reasoning, creativity, and memory, as well as providing a life history, mood, motivations, attitudes, and content. In particular, one of the most challenging parts of a VH is the interaction with a subject through a verbal conversation. The VH not only has to generate coherent, meaningful and contextualized messages, it also has to remember messages or ideas of the conversation. The origin of this type of interactive tools is found in chatbots and later in conversational agents.

The first software that allows to interact with natural language were chatbots. They are an informatic system that can establish a conversation with one or more users through different communication channels as voice, text or visual language (Abd-alrazaq, Alajlani, Alalwan, Bewick, Gardner, & Househ, 2019). However, classical chatbots have a pre-defined sequence of answers to the different possible inputs. It is a bounded system which response is already settled. The use of the chatbots is very diverse. The first operational chatbot is found in *ELIZA* (Weizenbaum, 1966). Since there, chatbots have been a very popular field of study. Many different algorithms have been used to improve the communication with a subject trying to overcome the past models. Some of them are *MegaHAL* (Hutchens & Alder, 1998) which is based in Markov's model basing its prediction in a probability distribution choosing between the most likely words for the answer. Chatbots finally evolved when artificial intelligent (AI) algorithms were incorporated to this field.

The use of intricate models capable of generating the most appropriate response to a given input has given rise to the concept of Conversational AI. Unlike chatbots, conversational AI algorithms are unbounded, with responses generated based on the input. These models undergo prior training with data from conversations, books, or other forms of text, enabling them to communicate with humans. Several strategies exist to augment the interaction with conversational AI and render it more human-like, primarily through the use of language models. These models have been employed to enhance the performance of chatbots by enabling them to generate more complex responses not merely extracted from a database, but based on them. Additionally, the work of Imam and Kodratoff (1997) defines an avatar as intelligent if it demonstrates proactive actions, responsiveness to the environment, and social interaction with other users. Early language models applied to generate conversations emerged from the work of Serban et al. (2016). The model developed in this study could perform two distinct tasks; providing technical support for Ubuntu programming and generating Twitter conversations. Today, there are various conversational AI technologies in use, such as Alexa, Siri, or Cortana (Hachman, 2014; Moemeka & Moemeka, 2015). GPT models by OpenAI (Brown, Mann, Ryder, Subbiah, Kaplan, Dhariwal, Neelakantan, Shyam, Sastry, Askell, Agarwal, Herbert-Voss, Krueger, Henighan, Child, Ramesh, Ziegler, Wu, Winter, Hesse, Chen, Sigler, Litwin, Gray, Chess, Clark, Berner, McCandlish, Radford, Sutskever, & Amodei, 2020) represents a significant advancement in this field, emerging as one of the most sophisticated language models to date. Given the type of training GPT underwent, it can introduce contexts that align with the model's response. This not only enables the delivery of generic yet precise messages, but it also allows for responses to be crafted according to certain guidelines. Consequently, it becomes possible to orchestrate a conversation wherein the AI can interact based on the context's instructions, such as generating extroverted or empathetic messages, thereby simulating personality traits. These AI models are nearing the ability to emulate a conversation in a typical social situation with another human (Burden & Savin-Baden, 2019). For instance, the study by Tack and Piech (2022) validates the use of GPT-3 and Blender as a question-answer teacher through written dialogues. However, despite these advancements, there is still a lack of research on emotion elicitation and recognition using conversational AI.

Furthermore, a significant portion of studies involving the use of conversational AI have traditionally relied on interaction with the subject through written text displayed on a computer screen. This

method of communication is somewhat unrealistic as it fails to replicate a real-life conversation between human beings. Currently, various AI models, such as voice synthesizers or audio transcription tools, could foster a more authentic conversation. Audio transcription models, which transcribe spoken language into written text, have been a focal point of recent research. Most of these models convert the audio input into a Mel Spectrogram to capture non-trivial features necessary for transcription. This task encounters several challenges, including the punctuation of sentences with periods or question marks, and transcribing different languages within the same sentence. However, novel Automatic Speech Recognition (ASR) models based on transformers such as Whisper are outperforming the task of transcribing voice (Radford et al., 2022). Conversely, voice synthesizer models have seen more extensive research over the years, yet certain issues remain unresolved, such as achieving naturalness in the synthetic voice. For a realistic conversation, it is crucial that the model's voice closely imitates a human voice. Additionally, appropriate intonation for questions or for conveying sadness or happiness is essential. The voice should also respect certain pauses, as indicated by periods or commas. This challenge has recently been addressed by platforms like AWS Polly, which incorporates commands in the plain text to indicate pauses or to instruct the model to speak louder or faster. The integration of VR and the aforementioned AI models could potentially construct a VH. Currently, most research in this area is conducted in 2D, using screens or smartphones (Angga, Fachri, Elevanita, Suryadi, & Agushinta, 2015). However, these mediums may not fully realize the potential degree of realism achievable with VR. The study by Ciechanowski, Przegalinska, Magnuski, and Gloor (2019) investigated the difference between a chatbot displayed through a screen interface and the same chatbot displayed through a terminal interface. The results revealed that an improperly displayed avatar could compromise the naturalness of the communication. Similarly, the work of DeVault et al. (2014) compared face-to-face human interviews with face-to-VH interviews in healthcare decision-making. This study yielded promising results, with VH interviews achieving ratings comparable to those of human interviewers. Interestingly, VH even outperformed human interviewers in terms of confidence ratings. Therefore, elements such as voice or appearance significantly influence the naturalness of the user's experience. VR could potentially enhance the sense of realism in a conversation with a VH and also allows interaction with and control of the VH within the virtual environment.

Nonetheless, achieving a realistic user experience with a VH presents a set of challenges that are currently difficult to overcome. The primary obstacles involve replicating the physical behaviors and movements of a human body in the avatar, in a bid to make the VH as realistic as possible. One of the significant challenges is lip synchronization, a task that only a handful of AI models are currently able to handle. Hand movements and gestures, which are vital for conveying naturalness, pose a similar hurdle. Another vital aspect is the ability of the avatar to evoke and express different emotions. This feature is not only critical from an emotion recognition standpoint, but it also necessitates a responsive avatar that can alter its facial expressions and gestures in response to the subject's messages. The study by Sollfrank et al. (2021) reveals that while artificial faces have different effects compared to real faces, both can stimulate a neurological response in the subject. Furthermore, Hube, Vidackovic, and Sedlmair (2022) demonstrates that realistic avatar faces can aid users in accurately identifying emotions more effectively than without them. Several solutions exist for off-streaming scenarios, but very few are available for real-time or live cases. In the realm of VR, there are platforms that enable the deployment of an avatar capable of synchronizing lip movements with an audio file. One of the most sophisticated and realistic platforms is Nvidia's Omniverse. It features a library called Audio2Face, which facilitates lip synchronization. Unity is also in the process of developing the Salsa library for coordinating lips and gestures. However, at present, no complete program or library can combine all the required elements to replicate the physical behavior of a human through the avatar of a VH in real time.

To the best of our knowledge, no previous work has developed a comprehensive VH incorporating a realistic avatar, conversational abilities based on Language Models, embedded psychological constructs such as personality, mood or attitudes, along with emotional facial expressions, lip synchronization, and voice synthesis, all intended for use in real-time semi-guided conversations. The primary objective of this research is to develop and validate a set of VHs that combine all these technologies and demonstrate the ability to elicit emotions in humans. To this end, we utilize various technologies from companies including AWS, Google, OpenAI, and Nvidia. Upon constructing the VH, all the different steps involved in the conversation are validated through an experimental setup involving 64 participants. This paper presents an analysis of the real-time performance of the conversational pipeline, an evaluation of human–machine interactions, the naturalness and realism of the generated conversation, as well as the emotions elicited in subjects, and the ability to identify VH's emotions. Our results reinforce the viability of utilizing VHs in the field of affective computing to induce emotions in ecologically valid contexts. Such advancements provide a range of applications in sectors such as health, education, and marketing, among others.

This paper is structured as follows: Section 2 elucidates the experimental design and the tools utilized therein, along with the analytical methods employed. Section 3 presents the results derived from the aforementioned analysis. Section 4 discusses the work accomplished and the objectives met, and provides a discussion of the results. Lastly, Section 5 concludes the study and highlights the key findings.

2. Materials and methods

2.1. Participants

A group of 64 subjects was recruited to participate in the experiment. The mean age of the subjects was 31.956, SD = 10.339 years, including 31 males, 32 females and 1 non-binary gender. The group was established based on specified inclusion parameters: (a) ages between 18 and 55 years, (b) right-hand dominance (due to possible influence on EEG patterns Galin, Ornstein, Herron, & Johnstone, 1982), (c) normal vision, and (d) Spanish as the first language, articulated with a Spanish accent. Exclusion conditions involved: (a) pregnancy or nursing, (b) presence of psychiatric disorders, and (c) usage of psychotropic drugs. To ensure the subjects constituted a homogeneous group and they were in a healthy mental state, they were filtered by the Patient Health Questionnaire (PHQ-9). PHQ-9 is a standard psychometric test used to quantify levels of depression. Significant levels of depression would have affected the emotional responses. Only participants with a score lower or equal than 9 were included in the study.

Prior to their participation, they received documentary information on the study and gave their written consent for their involvement. The responses were anonymized and randomized to ensure the privacy of the information. The study obtained the ethical approval of the Ethical Committee of the Polytechnic University of Valencia (P06_25_07_2022).

2.2. Instrumentation

The experimentation room was equipped with a 6.36×2 meters white screen, which was used as a method of projection for semi-immersive, life-size virtual reality (see Fig. 1). This screen was outfitted with two Optoma projectors (ZH400UST DLP), each with a resolution of 1920×1080 pixels, which after blending, provided a projection of 3600×1080 pixels. The room was also fitted with a Logitech 5.1 sound system, model z5500. The PC used has the following spec: CPU, Intel(R) Xeon(R) W-2265 CPU @ 3.50 GHz (24), RAM, 131585 MB, and GPU, Quadro RTX 6000/PCIe/SSE2.

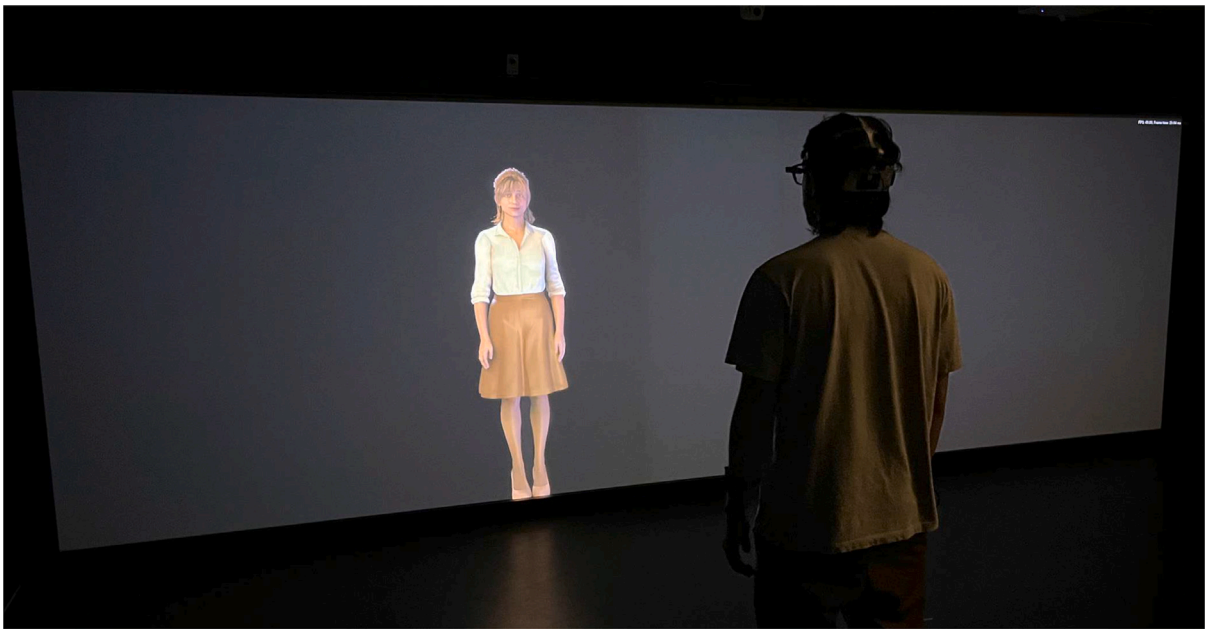


Fig. 1. Photo of the experimentation room.

During the experimental procedure, we collected a variety of data from each participant, encompassing speech, eye-tracking (ET), and physiological measures such as EEG, ECG, and EDA. Speech data was gathered using a SYNCO G1 A1 wireless microphone system, with the recording device positioned on the participant's forehead. ET information was recorded using the Pupil Invisible glasses from PupilLabs. The glasses feature a frontal camera with a resolution of 1088×1080 pixels, operating at a frequency of 30 Hz. The device recorded a the subject's gaze, and additionally provided an accelerometer and gyroscope, all functioning at 200 Hz. EEG and ECG signals were simultaneously captured utilizing the B-Alert x10 system (Advanced Brain Monitoring, Inc., USA), recording at a frequency of 256 Hz. EEG sensors were strategically placed in frontal (Fz, F3, F4), central (Cz, C3, C4), and parietal (POz, P3, P4) regions, adhering to the international 10–20 electrode placement system on the participants' scalps. A pair of reference electrodes was positioned below the mastoid. Electrode conductivity was confirmed with an aim to maintain electrode impedance below 20 kΩ. The ECG left lead was situated on the lowermost rib, while the right lead was placed on the right collarbone. Lastly, EDA signal was acquired using the Shimmer3 device, with a recording frequency of 128 Hz.

2.3. VH: general scheme

In our development of the VH, we employed a cognitive-inspired architectural design that divided the system into body and mind components. Drawing inspiration from the framework proposed by Burden and Savin-Baden (2019), the body component is further stratified into two subsystems encompassing senses and actions. A comprehensive overview of this design scheme is graphically represented in Fig. 2. The VH's body action subsystem shows an avatar that is visually rendered with facial expressions tailored to reflect its emotional state. As participants vocalize, their audio is relayed in real-time to the VH's senses subsystem. Upon the termination of speech, a real-time silence detector module identifies the cessation and proceeds to route the audio to a transcription module, converting speech into text. This transcribed text is then transferred to the VH's mind system. Here, a conversation module leverages a LLM to generate responses. These responses take into consideration various factors including the VH's life-history, the contextual situation, attitudinal disposition, prevailing mood, and motivations. Additionally, it references the conversation memory to ensure temporal coherence. The crafted response text is then passed to the action subsystem of the VH's body. This response is synthesized into an audio file, broadcast through speakers, while the VH avatar's lip movements are synchronized to mimic authentic speech. Concurrently, the senses subsystem is primed to anticipate the forthcoming participant response, marking the commencement of the next iteration of the process. During the experiment, the behavioral and physiological responses of the subject are collected to be modeled in future steps of the work.

2.4. VH body: actions

The objective of the Actions subsystem, which is a part of the body component, is to produce real-time audiovisual content for the VH, enabling real-time conversational interaction.

2.4.1. Avatar

Four realistic avatars were acquired, comprising one male and one female, each with two variations: one in a casual outfit and the other in a semi-formal outfit. Five alternative emotional expressions (neutral, happy, sad, relaxed, and angry) were developed for each avatar, yielding a total of 20 avatars. The size of the avatars were designed according with the height of the population in Spain, 1.76 m and 1.62 m for men and women respectively. After acquiring the avatars in FBX format, they were imported into Autodesk Maya 2022 for rigging and the generation of blendshapes. Blendshapes, which are 3D mesh deformers, totaled 122 for the female avatar and 125 for the male avatar. Using these blendshapes, the polygonal mesh of the avatar could be manipulated to generate specific expressions such as modifying the corners of the mouth or the furrowing of eyebrows for a sad expression. Blendshapes can be grouped based on prominent and common facial parts such as eyebrows, eyes, mouth, ears, nose, cheeks, and neck, all of which have a considerable number of blendshapes for necessary modifications. Eye blinking was also animated using blendshapes. Next, the avatar's body geometry was cut at the base of the neck, thus dividing the avatar's geometry into two parts: the head and the body. Following this modification, an idle animation was created for the

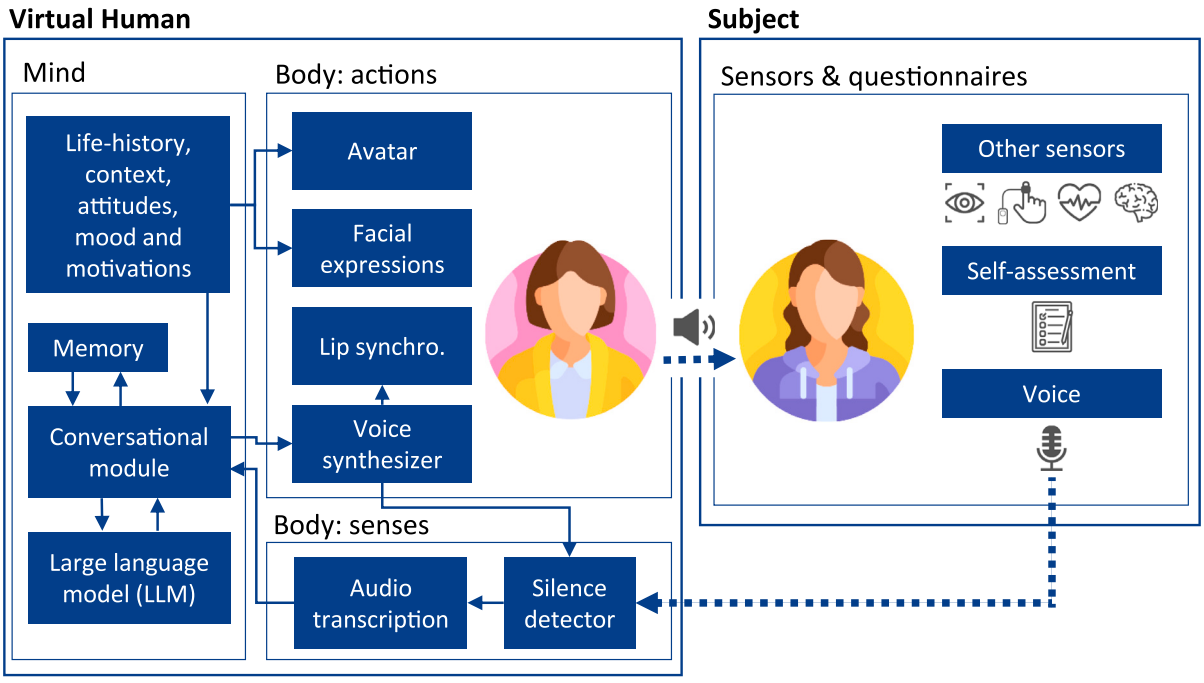


Fig. 2. Scheme of the VH's architecture and the human–machine interaction.

Table 1
Facial and body movements design per emotion.

Emotion	Arousal	Valence	Body movements	Mouth	Eyebrows	Nose	Extra
Neutral	Neutral	Neutral	Breathing	Neutral	Neutral	Neutral	–
Happy	High	Positive	Swaying	Moderate Smile	Slightly Raised Curved	Neutral	Open Eyes
Relaxed	Low	Positive	Breathing	Very Slight Smile	Slightly Raised Curved	Extended	Contracted Eyes
Angry	High	Negative	Swaying	Very Slight Inverted Smile	Lowered Straight	Contracted	Furrowed Brow
Sad	Low	Negative	Breathing	Slight Inverted Smile	Raised Brow	Neutral	Contracted Cheeks

avatar and, once complete, it was exported in USD format using an Omniverse connector plugin for Autodesk Maya (Legacy). We adopted the definition of 'avatar' as a visual representation that includes technical aspects such as geometry, blendshapes, textures, materials, and rigging. This avatar can be controlled either by a physical human or by a virtual human (VH), as described in Burden and Savin-Baden (2019). It is worth noting that some authors consider an avatar to be only those representations controlled by a physical human and may regard it as a component of a larger agent (Ospina-Bohórquez, Rodríguez-González, & Vergara-Rodríguez, 2022).

2.4.2. Facial expressions and body movement

The avatar's mouth, eyes, eyebrows, nose, and body movements were modulated based on the emotion to be elicited, utilizing the Facial Action Coding System (FACS) (Ekman & Friesen, 1978). The neutral avatar was designed with neutral features. The happy avatar was characterized by a moderate smile, a slight curved elevation of the eyebrows, open eyes, and a slight body sway. The relaxed avatar displayed a slight smile, a slight curved elevation of the eyebrows, a dilated nose, and contracted eyes. The angry avatar featured a very slight inverted smile, straight and lowered eyebrows, a contracted nose, and a furrowed brow accompanied by a body sway. The sad avatar exhibited a slight inverted smile, a raised brow, and contracted cheeks. All avatars presented a breathing animation. These facial expressions were designed based on the FACS and two online pre-tests were conducted with images of the faces on 102 subjects to refine the faces (further information in the supplementary materials). Table 1 show a summary of the traits for each mood.

2.4.3. Lip synchronization

Nvidia Omniverse's Audio2Face app was utilized for real-time lip synchronization. The previously created USD file with the idle animation was imported and the "Character Transfer" tool was used to animate the avatar's face. The Player Streaming feature was used to receive the synthesized VH voice. Texture modification could be achieved through the shader associated with each object's material, where the base color and opacity could be changed. To illuminate the scene, three rectangular lights were added: a rim light behind the character, tilted upwards with low intensity and a neutral color; another light at 45 degrees to the right of the character, tilted downward with high intensity and a warm color; and a third light at 45 degrees to the left of the character, tilted upwards with low intensity and a cold color. This lighting setup added realism to the character compared to the default lighting. The avatar was displayed directly in the full-screen mode of the application, prepared to synchronize the lips as the synthesized voice was played.

2.4.4. Voice synthesizer

The synthetic voice for the VH message was produced using Amazon Polly API, a service that leverages artificial intelligence models to convert text into an audio file in MP3 format. To provide a more natural and expressive prosody, we employed the Speech Synthesis Markup Language (SSML) for text input. This allowed us to introduce precise pauses at punctuation marks, such as periods and commas, with respective durations of 0.6 and 0.25 s. Two distinct voice identities were utilized: 'Lucia', a neural voice, provided the female articulation, while 'Enrique', a standard voice, was used for male avatars. Note that, at

Table 2
Summary of the different conversational parameters of each VH based on the mood.

	Neutral 1	Neutral 2	Angry	Happy	Sad	Relaxed
Life-history	Ana/David, 37 yo, Madrid, Pharmacist	Laura/Alejandro, 27 yo, Valladolid, Waiter/waitress	Marta/Jorge, 31 yo, León, Theater actor	María/Javier, 30 yo, Valencia, Teacher	Sara/Daniel, 25 yo, Ciudad Real, Studying psychology and retail associate	Lucía/Pablo, 21 yo, Bilbao, Book translator
Context	Subway stop	Tramway stop	Bus	Bar	Sitting on a park bench	Walking on the promenade
Motivation	–	–	She/He has been denied a salary increase and prices have gone up.	She/He has obtained a position as a primary school teacher in Valencia.	The landlord just called her/him to tell her/him that he needs the apartment in a month's time.	He/She is on vacation and has just left a spa.
Attitude	Kind, friendly and likes to talk about sport. Likes to chat and ask what sports the people she/he talks to like.	Kind, friendly and likes to talk about cinema. Likes to chat and ask what films the people she/he talks to like.	Needs to let off steam. She/He is not interested in the rest. Wants to talk about her/his anger.	Kind, friendly. Likes to talk and ask questions.	She/He does not feel like talking. She/He does not ask many questions.	Kind, friendly. Eager to talk and ask questions.
Greetings	Hello, my name is _ what is your name ?	Hello, my name is _ what is your name ?	My name is _ what is your name ?	Nice to meet you, my name is _ What is your name ?	Hello, my name is _ and yours ?	Hello, my name is _ what is your name ?
Farewall	I'm sorry, I have to go. We'll talk some other time. See you later!	I'm sorry, I have to go. We'll talk some other time. See you later!	Well, I have to go. Goodbye!	I'm sorry, I have to go. It was nice meeting you and talking with you. See you later!	I'm sorry, I have to go... See you later!	I'm sorry, I have to go. Nice talking with you. See you later!

the time of the experiment, AWS did not offer a neural voice for Spanish males. These selections were made based on their representative qualities, contributing to a more immersive interaction with the VH.

2.5. VH body: senses

The senses subsystem of the VH is designed to endow it with a capacity to perceive its environment, focusing specifically on auditory perception.

2.5.1. Silence detection

To facilitate real-time interaction with participants, we implemented an algorithm that detects the cessation of voice activity, signifying the end of a participant's speech. We utilized the silence detection model from the pyannote library (Bredin & Laurent, 2021), which ceases recording upon identifying a silence period exceeding one second. The onset and offset activation thresholds were both set at 0.5 s. The model commences silence checking three seconds after the onset of audio streaming. Should it identify an absence of voice activity lasting over a second, it ceases the streaming. Due to its operation in the background, this process does not cause any delay or interference with the participant's audio. The recorded audios are systematically stored in a designated folder for each participant and conversation.

2.5.2. Audio transcription

Participant speech is transcribed into text, in the speaker's native language, utilizing Google Cloud's Speech-to-Text service. This service not only transcribes spoken words but also includes punctuation marks such as periods and question marks, enhancing the clarity and interpretability of the transcribed message for subsequent analysis. The model accepts audio input in WAV format and provides an output featuring multiple components of the recorded audio. If it identifies distinct segments of speech as separate messages, it will return each message individually. A post-processing function was employed to consolidate all individual messages generated by the AI model, thus preventing any overlap of messages.

2.6. VH mind

The Mind system is designed to endow the VH with capabilities for natural language comprehension, rational thought, creativity, and memory. In addition, it furnishes the VH with a life history, context, attitudes, motivations, and mood.

2.6.1. Life-history, context, attitudes, mood and motivations

This module aims to create a narrative that provides a set of psychological features and context to the VH. In particular, it provides a life history (name, age, birthplace, current city, and profession), conversation's spatial context, motivation for the specific mood, and the VH attitude to engage in social interactions. The names used was selected from the list of most common names in Spain according to the Spanish National Statistics Institute, excluding compound names, trying to reduce any possible personal bias to a particular name. The current city was defined as Valencia (Spain) to provide spatial coherence, due to the experiment was performed in this city. The avatars with a non-neutral emotional state also include a motivation to engage in conversations on sports or cinema.

The constructed narratives conform to the following template: *"The following is a conversation with [name]. [name] is a [X]-year-old [nationality] [gender]. He/she was born in [birthplace] and currently lives in [current city]. [name] is [profession]. At the moment, [name] is [context]. Today, [name] is [mood] because [motivations]. Due to his/her [mood], [name] is [attitude]".*. Given that this module establishes the VH's gender and mood state, this information is employed to manually select an avatar

Table 3
Illustration of the prompt used to create a VH's response.

"The following is a conversation with Marta. Marta is a 31 year old Spanish woman. Marta was born in León and is living in Valencia. Marta works as a theater actress. Marta is on a bus and is talking to the person sitting next to her. Today Marta is angry because she has asked for a salary increase and has not been granted it. This year the prices of electricity, rent and food have gone up and with Marta's salary it is difficult to make ends meet. In this situation, Marta cannot spend money on activities she likes to do in her free time, such as going to the gym or to the movies, because she has to save money to be able to pay for all the expenses. Today Marta is angry and needs to let off steam. That's why Marta is not interested in getting to know the people she talks to and only wants to tell them why she is angry. Marta: My name is Marta. What is your name? (SUBJECT'S NAME): (...) (CONVERSATION LOG) Marta: "

that coheres with these traits. To evade repetitiveness in opening and concluding phrases, each avatar is imbued with a distinct greeting and farewell. We hypothesize that the pre-defined mood embedded in the VH is capable of triggering the VH's emotions during the conversation and can elicit emotions in the subjects.

Table 2 furnishes a summary of the parameters defined in this module for each VH.

2.6.2. Conversational module

The Conversational Module plays a crucial role in crafting the prompt that is sent to the large language model LLM. It integrates a variety of inputs, namely the emotional narrative (encompassing life-history, context, attitudes, mood, and motivation), the most recent transcription of the subject's audio, and, when applicable, the log of the conversation from the memory module. Notably, all these inputs are in Spanish. To personalize the dialogue template, the subject's name is manually added at the onset of the experiment. Using this information, a structured and context-sensitive prompt is assembled following the defined sequence. Subsequently, this prompt is transmitted to the LLM for processing. Interestingly, the LLM occasionally generates not just the VH's response but also anticipates the subsequent interaction from the participant due to the number of token to generate is fixed. Hence, the module performs a post-processing step to filter out a single coherent response from the VH. Additionally, this module makes adjustments for specific words or expressions that the synthesis process struggles to handle accurately, such as "jaja" or "(laughs)". Table 3 presents an illustrative example of the prompt crafted by the Conversational Module:

2.6.3. Large language model

The LLM employed in this study is GPT-3 by OpenAI, specifically utilizing the *text-davinci-002* variant via API. The model had different parameters such as temperature, presence penalty, frequency penalty and maximum number of tokens which were meticulously calibrated. Temperature determines the output's level of focus and determinism; higher values make the output less deterministic. This parameter ranges from 0 to 1, and it was set to 0.9 in this study. The presence penalty parameter controls the introduction of new tokens and enhances the model's reluctance to discuss novel topics as the parameter value increases. Ranging from −2 to 2, the presence penalty was set to 0.9. The frequency penalty penalizes new tokens based on their existing frequency in the text, reducing the likelihood of the model repeating the same line verbatim. This parameter also varies between −2 and 2, and it was set to 0. Lastly, the maximum number of tokens indicates the quantity of new tokens generated in the language model's response. It can range from 1 up to the model's token size limit, which was 4000 in this case. For this study, the maximum number of tokens was set to 150. These parameter settings were established through an exhaustive iterative process involving numerous trials during the pre-pilot phase.

2.6.4. Memory

The memory module serves a pivotal role in preserving the conversation's log. This stored log is subsequently utilized by the Conversational Module to uphold the short-term temporal coherence, thereby ensuring the continuity and flow of the conversation.

2.7. Experimental procedure

In the initial phase of the experiment, participants were asked to fill out a battery of psychological questionnaires encompassing demographic information, PHQ-9, State-Trait Anxiety Inventory Questionnaire (STAI), and several other physiological assessment questionnaires, which will be used for further analysis. STAI was used to assess the emotional baseline of each participant. Following this, participants were equipped with the sensors described and proceeded to engage in six distinct conversations with a set of six VHs. They were displayed at a realistic scale via a projector encompassing an entire screen in a dimly-lit room, where the screen served as the primary source of light. This configuration helped in concentrating participants' attention on their communication with the VHs. The commencement of each interaction took place at a distance of 2.60 meters from the screen, although adjustments for cognitive and social comfort were allowed. The research team oversaw and managed the experiment from a separate location to minimize any potential influence on participants' responses.

The experimental procedure is shown in Fig. 3. The initial two interactions involved neutral-emotion VHs, with the order of gender representation counterbalanced. Subsequently, participants engaged with four emotionally expressive VHs expressing happiness, relaxation, sadness, and anger, with the order and gender of these interactions again counterbalanced. Prior to each interaction, participants were provided with a situational context (e.g., "You are at a tramway stop and a person initiates a conversation with you"). The VH initiated the dialogue with a greeting, enabling participants to partake in open-ended conversations absent of specific guidelines or objectives. Interactions could be concluded by participants bidding farewell to the VH, prompting the experimenter to manually terminate the task. Alternatively, after a maximum duration of 4 min, the VH automatically bid goodbye, and the task was concluded by the experimenter.

Following each conversation, participants were instructed to assess both the naturalness and realism of the social interaction. Naturalness is appraised in terms of the conversation's flow, spanning from "I felt very forced during the conversation" to "I felt very natural during the conversation". Realism assesses the content generated during the conversation, ranging from "It was an artificial conversation. It doesn't resemble a real conversation at all" to "It was a realistic conversation. The content was very similar to that of a real conversation". Both metrics employ a 7-point Likert scale.

In addition, they were asked to identify both the VH's emotions and their own, utilizing the Self-assessment Manikin (SAM) questionnaire (Lang, 1980). This is a standardized measure of valence and

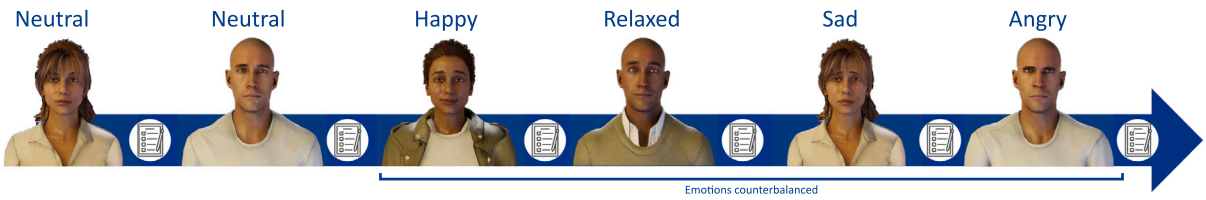


Fig. 3. Scheme of the experimentation protocol.

arousal referring to Russell's Circumplex Model. In this context, participants were asked to rate both subjective and VH's valence and arousal using two 9-point Likert scales during their conversations. We hypothesize that a happy VH will induce high arousal and positive valence; relaxed VH, low arousal and positive valence; sad VH, low arousal and negative valence; and angry VH, high arousal and negative valence. We are setting 5 as the threshold on both scales, as it represents the midpoint of the Likert scale. Data collection was executed from October 2022 to February 2023.

2.8. Data analysis

The present analysis of the experiment is structured into five distinct sections. The first section focuses on evaluating the time-processing efficiency of the pipeline employed, paying particular attention to the time expended by the modules that incorporates AI-based APIs: audio transcription, LLM, and voice synthesizer. The pipeline log, based on UNIX time, was used for this computation, assessing the interval between input receipt and output provision by each module. The remaining pipeline modules, those not reliant on external AI-based API services, were collectively evaluated.

The second section embarks on a descriptive analysis of human–computer speech interaction, drawing upon features extracted from the conversations. For clarity, we define a 'conversation' as the total time frame from greeting to farewell. An 'interaction' refers to a cluster of one or more consecutive phrases that are conveyed concurrently within the same audio recording or synthesized by either the participant or the VH respectively. A 'sentence', on the other hand, pertains to a distinct phrase with an identified or fabricated endpoint. Our analysis covered the total duration of the conversation, the number of interactions per conversation, the number of sentences per conversation, and the number of words per conversation. Additionally, we evaluated the total duration and speaking duration per interaction. For human participants, interactions were dissected into three phases: (1) 'Time to start talking', representing the silent interval between the initiation of the recording and the beginning of speech, (2) 'Time talking per interaction', and (3) 'Time of final silence', the span between when participants stopped speaking and the conclusion of the recording. Silent periods were identified using pyannote voice activity detection model (Bredin & Laurent, 2021). It should be noted that as the VH's voice is synthesized, there are no segments pertaining to 'Time to start talking' or 'Time of final silence'; hence, the total and speaking durations are identical in this instance. Lastly, the interaction time was further evaluated in relation to its position within the conversation for both the human interlocutor and the VH. This computation subsequently differentiated between the initial audio of the human and VH, the second audio, and so forth. Fig. 7 clearly illustrates the average evolution of speaking duration throughout the conversation. Due to non-normality of the data based on Shapiro–Wilk test, Mann–Whitney-Wilcoxon test were applied to analyze differences between VH and subjects interactions.

As the final component of our analysis of human–computer speech interactions, we conducted a manual review to assess instances where errors in the interaction could be produced. These errors are spetarated in two, depending on the source, which are the silence detection module and the LLM exhibited errors. Errors arising from the silence detection pipeline were quantified by tallying the number of times in which the VH interrupted the subject during the subject's speech within each conversation. Conversely, errors stemming from incoherent responses generated by the LLM encompassed various categories, including repeated messages, statements made in the third person, messages that were inconsistent with prior conversation topics, the use of incorrect words, including foreign or nonexistent terms, and technical errors, such as instances where the VH failed to comprehend the human's sentence.

Naturalness and the realism of the conversations was analyzed for each type of VH.A Friedman test is used to identify statistical differences among various VH conditions, averaging both neutral scores. This is followed by a post-hoc analysis using the Nemenyi test with Bonferroni correction to examine statistical differences between distinct states. Additionally, a Wilcoxon signed-rank test is employed to evaluate the average scores in neutral versus emotional conditions for each subject.

Finally, the emotional responses elicited in participants (i.e., the emotions experienced by the subjects), and the identification of the emotional state of the VH (i.e., the emotion the participant perceives the VH to be experiencing) have been explored in Sections 3.4 and 3.5. The results were collected based on valence and arousal utilizing the Self-Assessment Manikin (SAM) scale (Bradley & Lang, 1994), employing a 9-point Likert scale. Once the normality of the distributions is verified, a repeated measures ANOVA is performed over the different emotional states. A post-hoc analysis with Bonferroni correction is performed to analyze statistical differences between the different emotional states. One subject was excluded from the analyses of naturalness, realism, emotion elicitation, and emotion identification because his/her score on the STAI questionnaire surpassed the 95th percentile (47 for women and 40–41 for men) (Guillén-Riquelme & Casal, 2011).

3. Results

3.1. Technical performance

Fig. 4 presents the distribution of processing times utilized by each module to generate an output. The LLM, specifically OpenAI's GPT-3, was the most time-consuming module with a mean processing time of 1.82 ± 1.46 s, constituting 41.07% of the total conversation pipeline duration. The 99th percentile of the distribution is at 6.89 s. The audio transcription module, employing Google's Speech-to-Text technology, also demonstrated a substantial processing duration of 1.70 ± 0.62 s, corresponding to 38.14% of the total pipeline time. In comparison, the voice synthesizer module, powered by AWS Polly, required a relatively minor average processing time of 0.09 ± 0.04 s to complete the voice synthesis, amounting to only 2.13% of the pipeline's overall time. The remaining modules, encompassing data recording and storage as well as text cleaning, required an average processing time of 0.82 ± 0.42 s, equating to 18.65% of the total pipeline time. Overall, the entire conversation pipeline commanded an average processing time of 4.44 ± 1.77 s, where the 99th percentile of the time consumption is located in 10.18 s.

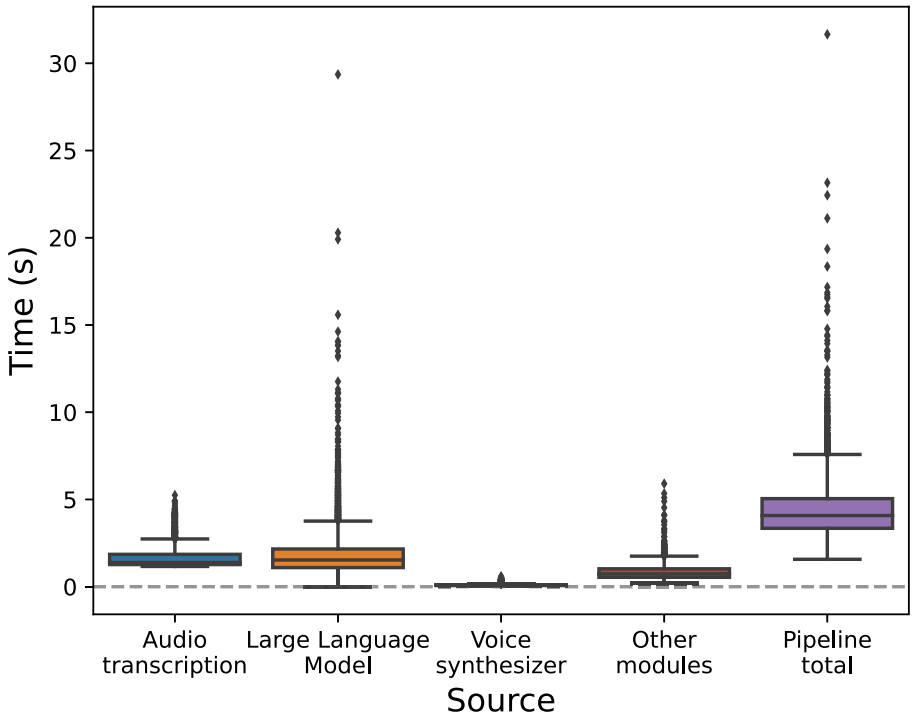


Fig. 4. Boxplot of the time spent by the whole pipeline and splitted by each module.

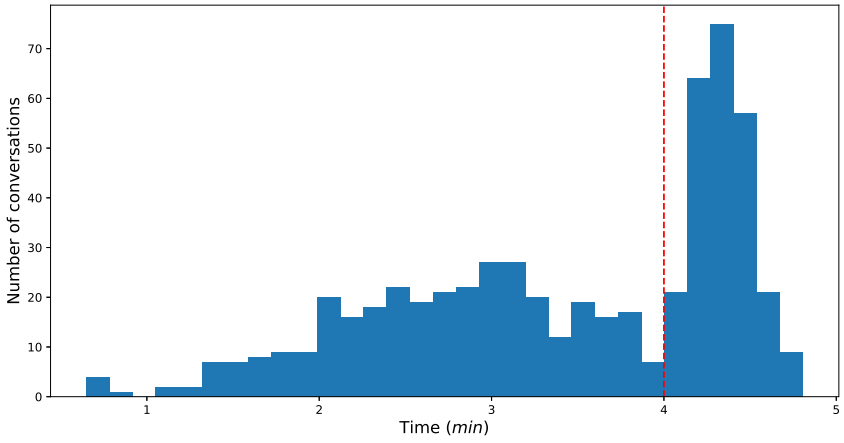


Fig. 5. Distribution of the time conversation duration for all conversations.

3.2. Human–machine interaction

The average duration of a conversation is 3.44 ± 0.97 min, involving an average of 20.81 ± 6.51 interactions per conversation between the VH and the participant. Fig. 5 shows the time distribution of the conversation durations. It illustrates two distinct distributions, separated by a temporal threshold set at 4 min to conclude the conversations. Conversations concluding below this threshold exhibit an average duration of 2.730 ± 0.712 min and includes 57.34% of the conversations. Conversely, the second distribution encompasses conversations that exceed the 4 min threshold, terminating with an average time of 4.349 ± 0.163 min. Notably, 42.66% of conversations extend beyond this threshold.

The mean number of sentences used by the subject was 17.23 ± 5.60 sentences per conversation. In contrast, the VH uses significantly more sentences, averaging 23.71 ± 6.97. According to the Mann–Whitney test, there is a statistically significant difference ($p < 0.001$) between these two distributions. As for the word count, a conversation involves an average of 112.46 ± 54.85 words contributed by the human participant, while the VH uses more words, averaging 145.25 ± 51.46. Again, the Mann–Whitney test reveals a statistically significant difference ($p < 0.001$) between these distributions.

With respect to audio interaction features, the extracted results are outlined below and visualized schematically in Fig. 6. The human participant typically takes an average of 1.27 ± 1.91 s to begin speaking, with their speaking duration averaging 3.94 ± 1.61 s. The VH, conversely, engages in lengthier speaking durations, averaging 5.74 ± 3.85 s. A statistically significant difference is indicated by the Mann–Whitney test ($p < 0.001$). The silence detection pipeline's average time to cease recording following the participant's discontinuation of speech is 2.80 ± 0.54 s. The average audio duration for the human participant is 8.01 ± 3.66 s, while the VH's average audio duration is slightly lower, at 5.74 ± 3.85 s, presenting statistical differences ($p < 0.001$). The audio duration and the end of the VH time speaking is the same because the API of the module returns the audio with only the sentence of the VH. The use of the silence detection algorithm is not necessary.

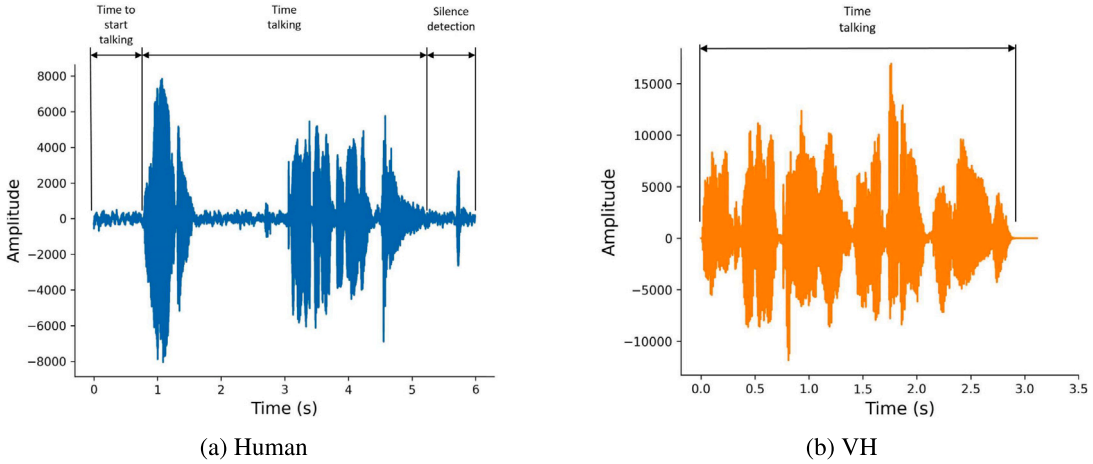


Fig. 6. Schematic of the audio interaction features extracted for human and VH.

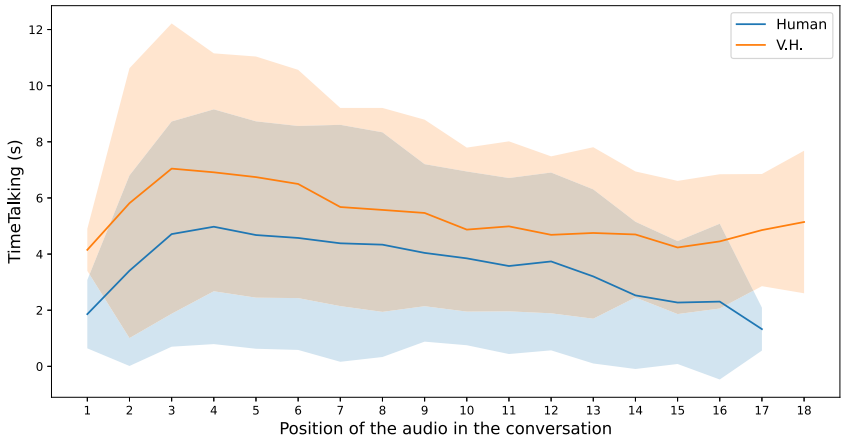


Fig. 7. Mean duration of the audios in terms of each position in the conversation for the Human and the VH. The average time is represented by a line whereas the standard deviation of the duration is the shadowed area.

Fig. 7 illustrates the average speaking duration per audio segment, categorized by their position within the conversation for both the human participant and the VH. On average, the VH spends more time speaking than the human. However, similar trajectories are observed for both, as the initial and concluding audio segments tend to be shorter, while those in the third and fourth positions are typically longer.

Table 4 presents the percentages of errors per conversation and the corresponding impact on individual interactions arising from various error types, as well as the cumulative effect across all conversations and interactions. Among these error categories, the most prevalent error source stems from VH interruptions induced by the silence detection algorithm, affecting 1.22% of individual subject interactions, which form part of a total of 11.24% conversations. Concerning the LLM, inconsistent messages are the most common error affecting the 0.85% of the interaction, that form part of the 9.12% of the conversations. It is follow by technical errors, repeated message, 3rd person message and incorrect words. A total of 2.62% of samples and the 23.28% of conversations have been affected by any different error from the LLM. In summary, the 3.84% of the interaction show some type of error, affecting 29.02% of the conversations.

3.3. Naturalness and realism

The perceived naturalness (pertaining to conversation flow) and realism (related to conversation content) are analyzed based on the emotional states of the VHs. The averaged scores for each VH, along with the statistical values from the Friedman test across different groups, are presented in Table 5. This analysis reveals distinct differences in both naturalness and realism across various VHs. Post-hoc analysis, however, does not reveal any statistically significant differences between the four non-neutral emotions (relaxed, happy, sad, and angry). Yet, these emotions collectively exhibit statistical differences when compared to the neutral state in all cases, with the exception of the relaxed-neutral comparison in terms of realism. To further analyze these findings, we examine the statistical differences between the emotional and neutral states of the VHs. The Wilcoxon signed rank test reveal a statistically significant difference between both groups, with emotional states outperforming neutral ones by approximately one point in both naturalness and realism. The overall scores for naturalness and realism are 4.6 and 4.4 respectively, placing them above the midpoint on the Likert scale from 1 to 7, suggesting a general sense of satisfactory realism and flow in the conversations with the VHs.

3.4. Emotion elicitation

Table 6 presents the averaged score of participant self-assessment using SAM for valence and arousal using the Likert scale from 1 to 9. These scores describes the VH's ability to elicit emotions in subjects. Repeated measures ANOVA reveals statistical differences between conditions in terms of valence, but not in terms of arousal. As illustrated in Fig. 8, the positions of both the angry and relaxed VHs align with their respective theoretical emotion quadrants according to Russell (1980). However, the VH exhibiting happiness induces less arousal

Table 4
Percentage of conversations and interactions affected by each type of error.

Source	Error type	Affected interactions (%)	Affected conversations (%)
Silence detector	VH interruption	1.22	11.24
LLM	Repeated message	0.59	6.35
	3th Person message	0.33	3.59
	Inconsistent message	0.85	9.12
	Incorrect words	0.08	0.83
	Technical errors	0.77	8.29
	Total	2.62	23.28
Total affected		3.84	29.02

Table 5
Average subject assessment of the VHs naturalness and realism, standard deviation in parenthesis.

Personality	Naturalness (1–7)	Realism (1–7)
Neutral	4.16 (1.32)	4.20 (1.38)
Angry	4.64 (1.70)	4.75 (1.70)
Happy	5.00 (1.58)	4.74 (1.65)
Sad	5.02 (1.72)	4.89 (1.60)
Relaxed	4.90 (1.75)	4.64 (1.45)
p-value	<0.001 (***)	<0.001 (***)
Neutral	4.16 (1.32)	4.20 (1.38)
Emotional	4.89 (1.39)	4.75 (1.30)
p-value	<0.001 (***)	<0.001 (***)
All	4.52 (1.41)	4.48 (1.37)

Table 6
Scores of the valence and arousal for subject self-assessment. Standard deviation is inside parenthesis.

Personality	Valence (1–9)	Arousal (1–9)
Angry	4.89 (1.89)	5.23 (1.72)
Happy	6.62 (1.90)	4.66 (2.02)
Sad	5.39 (1.93)	4.69 (1.79)
Relaxed	6.16 (1.89)	4.74 (1.67)
p-value	<0.001 (***)	0.262

(*) *p*-value < 0.05, (**) *p*-value < 0.01, (***) *p*-value < 0.001.

Table 7
Statistical comparison between the different emotional states for subject self-assessment over the different VH personalities.

Personality		*p*-value	
		Valence	Arousal
Angry	Happy	<0.001 (***)	0.078
Angry	Sad	0.118	0.053
Angry	Relaxed	<0.001 (***)	0.060
Happy	Sad	<0.001 (***)	0.890
Happy	Relaxed	0.148	0.715
Sad	Relaxed	0.014 (*)	0.827

Table 8
Scores of the valence and arousal for subject identification. Standard deviation is inside parenthesis.

Personality	Valence (1–9)	Arousal (1–9)
Angry	4.98 (2.15)	6.15 (1.90)
Happy	6.87 (1.54)	4.91 (1.83)
Sad	5.39 (2.29)	5.08 (1.85)
Relaxed	6.51 (1.54)	4.16 (1.66)
p-value	<0.001 (***)	<0.001 (***)

(*) *p*-value < 0.05, (**) *p*-value < 0.01, (***) *p*-value < 0.001.

than expected, resulting in its placement outside of its corresponding quadrant. Conversely, the VH expressing sadness elicits a greater degree of valence than hypothesized, leading to its repositioning into the relaxed quadrant. A pairwise comparison was undertaken to examine the distribution of arousal and valence experienced by participants after

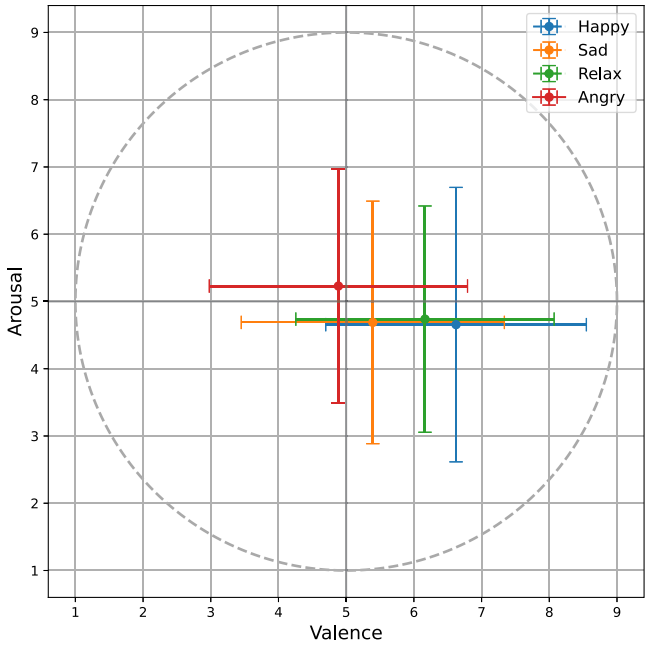


Fig. 8. SAM questionnaire results for subject self-assessment. The dot shows the average value of the valence and arousal. The vertical and horizontal lines show the standard deviation in terms of arousal and valence for each different emotion.

interacting with the VH, depending on each distinct emotional state of the VH. The results of post-hoc analysis are displayed in Table 7. In regard to valence, the hypothesized differences manifested in pairwise assessments. These include contrasting anger and happiness, anger and relaxation, along with happiness and sadness. However, the comparison between sadness and relaxation was the sole exception, displaying no notable differences. On the other hand, no statistical differences were identified in terms of arousal elicited by the different emotional states of the VHs.

3.5. Emotion identification

The average score for subject emotion identification of the VH is shown in Table 8. The results of the repeated measures ANOVA show statistical differences in valence and arousal. Fig. 9 shows that the location of the relaxed VH fits the theoretical model of arousal and valence (Russell, 1980). Angry VH is very close to its theoretical quadrant, with a high arousal and a valence close to the scale neutral point. Similarly, happy VH is very close to its corresponding with a positive valence and an arousal close to the scale neutral point. Finally, the sad VH gets slightly more arousal and valence than expected, achieving the position in the quadrant correspondent to the happy VH. Table 9 includes the post-hoc analysis. Regarding valence, the study successfully demonstrated all the hypothesized differences, specifically between the emotional states of anger and happiness, anger and relaxation, happiness and sadness, as well as sadness and relaxation. As

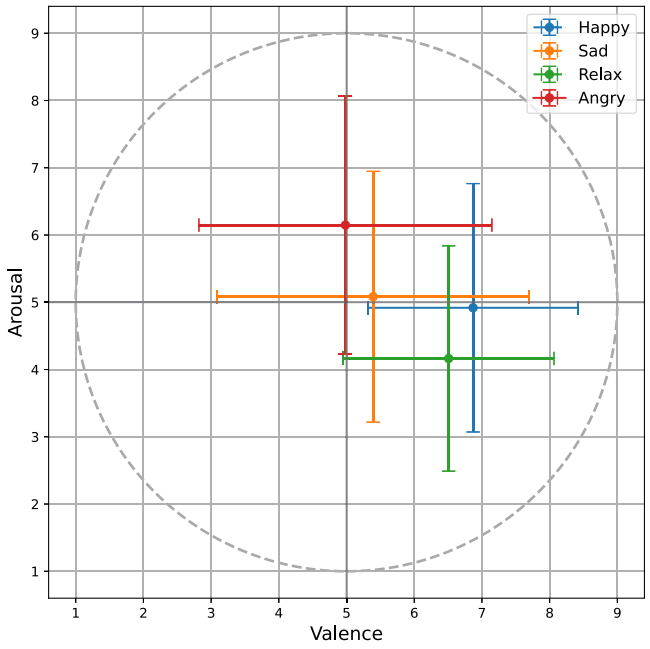


Fig. 9. SAM questionnaire results for subject identification. The dot shows the average value of the valence and arousal. The vertical and horizontal lines show the standard deviation in terms of arousal and valence for each different emotion.

Table 9
Statistical comparison between the different emotional states for subject self-assessment over the different VH personalities.

Personality		*p*-value	
		Valence	Arousal
Angry	Happy	<0.001 (***)	<0.001 (***)
Angry	Sad	0.285	0.001 (**)
Angry	Relaxed	<0.001 (***)	<0.001 (***)
Happy	Sad	<0.001 (***)	0.493
Happy	Relaxed	0.119	0.002 (**)
Sad	Relaxed	<0.001 (***)	0.004 (**)

(*) p-value < 0.05, (**) p-value < 0.01, (***) p-value < 0.001.

for arousal, not only did the study confirm hypothesized contrasts, such as those between anger and sadness, and anger and relaxation, it also unveiled unanticipated differences, namely between anger and happiness, and sadness and relaxation. However, the results did not corroborate hypothesized distinctions in the case of happiness versus sadness and happiness versus relaxation.

4. Discussion

This work introduces the first comprehensive VH system specifically designed for real-time emotion elicitation during semi-guided conversations. It was created through the integration of cutting-edge AI and VR platforms, based on a cognitively-inspired architecture where each module aims to perform a particular task in a mind/body scheme. The primary objective of this research was to validate the developed VHs as emotional stimuli, exploring the range of emotions evoked in the subject during their interaction with the VH. The generation of the emotions was based on modifications in VH's body (i.e. facial expressions) and the mind (i.e., attitudes, moods, and motivations represented in the prompts sent to the LLM). For this purpose, a total of twenty VHs were crafted, having five emotional states (neutral, anger, happiness, sadness, and relaxation), two genders (male and female), and two types of appearance (casual and semi-formal).

4.1. Virtual Human architecture

This research introduces a novel framework for developing VH, founded on the idea of bifurcating modules into mind and body, a concept adapted from the architecture proposed by Burden and Savin-Baden (2019). The body system is further delineated into action and sensory subsystems. The action subsystem is engineered to generate real-time audio-visual content for the VH, thereby facilitating real-time interactive conversations. This involves the integration of a lifelike avatar completed with facial expressions, synthesized voice, and lip synchronization. Notably, we leveraged AI-based models: AWS Polly API for instantaneous voice synthesis, and Nvidia audio2face for real-time lip synchronization. This approach culminates in a realistic audio-visual interface that is projected on a large screen, displaying a life-size avatar. The sensory subsystem equips the VH with auditory capabilities. It comprises a silence detection module, which conducts ongoing analysis to ascertain when the interlocutor has finished speaking, and an audio transcription module based on Google's AI model. This module provides the cognitive competency to interpret the spoken audio. The mind system furnishes the VH with psychological traits along with cognitive and emotional aptitudes. Specifically, we endowed it with cognitive proficiencies such as language comprehension, rational thought, creativity, and memory, utilizing GPT-3. Additionally, we incorporated psychological traits and states such as life-history, context, attitudes, moods, and motivations into the prompts sent to the LLM. To ensure temporal coherence, we integrated a memory module. The participants were equipped with a microphone enabling them to converse with the VH. Further, data pertaining to ET, EEG, ECG, and EDA were collected. These biometric data will fuel future research, facilitating the development of automatic emotion recognition systems and biofeedback intelligence. Several recent studies have deliberated on approaches to create VHs. For instance, Miao, Kozlenkova, Wang, Xie, and Palmatier (2022) proposed a four-class taxonomy rooted in the degree of form and behavioral realism. A 'Digital Human' necessitates a realistic anthropomorphic appearance, intelligence, autonomous and natural verbal and non-verbal communication, cognitive, affective, and social abilities. Our present architecture attempts to encompass all these elements, thereby illuminating the path for future research in the evolution of VH.

4.2. Technical performance

The architecture under study has been technically dissected, with time expenditure for each module thoroughly analyzed. Initial focus was directed towards three modules that incorporated AI-based API services, namely the voice synthesizer, the audio transcription, and the LLM. Remaining modules were evaluated collectively. The LLM module emerged as the most time-consuming, with an average time expenditure of 1.826 ± 1.462 s (41.07%). While alternative LLMs offering greater efficiency are available, GPT-3 was chosen for its superior conversation quality and coherence compared to other text generator models. Moreover, the usage of GPT-3 through OpenAI's API bypasses the need for local computation of model responses, thus preventing potential tool slowdowns. At the onset of data collection, GPT-4 was not available. Importantly, the selected model should balance between high-quality text generation and time efficiency to maintain conversational naturalness. Slow models may compromise this aspect. The second most time-consuming module is the audio transcription module, consuming an average time of 1.70 ± 0.62 s (38.14%). The Whisper ASR model was considered but was ruled out due to its local computation requirement and time-intensive nature. The voice synthesizer module operates the fastest AI-based API for generating the VH's voice, requiring 0.09 ± 0.04 s (2.13%). Although synthetic speech generation models are emerging in the open-source realm, as far our knowledge, the AWS's speed currently outperforms those deployed locally. As new AI models continue to evolve and garner interest, it is anticipated that more accurate and

efficient models will be developed and become accessible. Currently, reliance on external AI-based APIs is essential to yield high-quality VHs, primarily due to (1) the scarcity of open-source AI models for translate and synthesize voices, as well as generate text, and (2) the high computational resources demanded by such AI models, necessitating large-scale computing distributions. Nevertheless, the modular nature of the proposed VH architecture facilitates the independent testing and time optimization of new AI models, enhancing each module individually. The remaining modules collectively consume 0.82 s, accounting for 18.650% of the total communicative pipeline. Despite its relatively small proportion, this time duration could potentially be curtailed. Further exploration into alternative data processing or saving mechanisms could reduce this duration, thus contributing to a more fluid conversation.

4.3. Human–machine interaction

Descriptive features were extracted from audio recordings of each conversation, providing a quantitative evaluation of duration and the nature of verbal human–machine interactions between the subject and the VH. The conversation's average duration amounted to 3.38 ± 0.98 min, encompassing an average of 20.61 ± 6.40 interactions. However, the histogram (Fig. 5) clearly displays two normal distributions with distinct central tendencies. 57.34% of the conversations belong to the distribution that does not exceed the 4 min threshold, with a mean of 2.730 min. Notably, 42.66% of the conversations exceed the threshold. This indicates high engagement experienced by the subject during the conversation and suggests a considerable amount of information condensed within the conversations.

On the other hand, statistical analysis unveiled that the VH consistently employed on average more sentences (23.71 vs 17.23, +37%) and words (144 vs 110, +30%) than human participants, culminating in extended periods of VH's speech (5.74 vs 3.94 s, +45%). These results indicate a higher level of dominance by the VH in the conversation, likely attributable to its design focus on triggering emotional narratives. Given this, the engagement appears quite balanced, with users demonstrating a high level of involvement during the conversations, as evidenced by sentence and word counts. Additionally, the duration of interactions suggests that the conversations were notably fluid. Maintaining such fluidity and balance was crucial in this experimental protocol to prevent excessive dominance by either the VH or the subject in conversations. It is pertinent to note that while subjects' speech rates may vary, the VH maintains a fixed speech rate which could differ from that of the subjects. Future research will examine variances in semantic and prosodic speech features. Furthermore, the pursuit of developing more sophisticated VHs that can closely mimic human speech remains an ongoing challenge.

The audio derived from the interaction with the subject can be segmented into three parts: (1) time to start speaking, (2) speaking duration, and (3) silence detection. Subjects, on average, commenced speaking 1.27 s after recording initiation. This part represents a natural waiting time until the subject detects that the VH stops speaking. However, the silence detection module holds paramount importance within the pipeline, tasked with the precise identification of dialogue termination or silence within audio streaming. However, a potent AI model capable of detecting the end of a dialogue in streaming audio is yet to be realized. To bridge this gap, we implemented a silence detection algorithm based on audio activity detection, facilitating uninterrupted dialogues without preset time thresholds for speech pauses. Nevertheless, the algorithm necessitates audio recording each time it is invoked, potentially causing considerable conversation delays. The average time spent during an interaction loop, identifying when a user concludes speaking, is measured to be 2.808 ± 0.546 s. This duration is amenable to optimization for delay reduction. Our algorithm employs diverse time thresholds to safeguard against information loss. Humans can quickly tell when someone has finished speaking due to the rhythm and tone of speech, or what is known as prosody. This suggests that upcoming silence detection systems should use these prosodic features to more quickly detect when speech has ended instead of a fixed threshold of activity detection. It should be noted that the pipeline duration, from the moment the silence detection module concludes that the subject has ceased speaking until the VH begins its response, averages at 4.445 s (Fig. 4). Comparatively, the real-time silence detection requires 2.808 s. Thus, a substantial portion of the waiting period for the subject, prior to receiving a response, is dedicated to ascertaining that the subject has completed their dialogue.

The errors arising from human–machine interaction were manually reviewed post-experimentation and categorized into distinct types, such as VH interruptions, repetitive VH messages, and incorrect word usage, with the corresponding percentages of affected interactions and conversations detailed in Table 4. Although the silence detector interrupts only 1.22% of the interactions, it can disrupt the flow of 11.24% of them. Consequently, it is a critical module slated for future enhancements, incorporating a more sophisticated model considering not just voice activity but also prosodic dynamics to discern if the subject has concluded the interaction. In contrast, the LLM issues affect between 0.08% and 0.85% of interactions, impacting a total of 23.28% of conversations. Utilizing future versions of LLM will ameliorate these ratios. Nonetheless, it is crucial to highlight that while these issues impede a specific interaction during the conversation, they do not necessarily affect the entire conversation. The subjects' perception of the conversation flow is analyzed in the subsequent subsection.

4.4. Naturalness and realism

The naturalness and realism of the model has an average score of 4.52 and 4.48 over 7 correspondingly. The highest score in naturalness and realism is achieved by the sad VH which is 5.02 and 4.89 respectively. On the other hand, it can be seen how the neutral VH achieves, in average, lower punctuation's in both scores. This could be due to different reasons. The first one is that emotion increases the sense of presence, as stated in previous literature (Diemer, Alpers, Peperkorn, Shiban, & Mühlberger, 2015; Riva et al., 2007). Therefore, as long as the emotion is expressed by the VH, the engagement and social presence of the subject increases, enhancing the naturalness and realism of the conversations. This results present a new step in analyzing the relationship between presence and emotions during conversations with VH, supporting previous evidence. However, we need to consider that the neutral VHs are the ones that are seeing firstly by the subject, since they were used as an adaptation process, allowing subjects to familiarize with the system. As the experimentation progress, they get used to the look and the interaction with the VH due to their exposure, and the scores of realism and naturalness may increase.

4.5. Emotion elicitation

The results of the SAM regarding the self-assessment showed differences in terms of the valence elicited in the subjects by VHs (Table 6). In general, there is a VH's tendency of eliciting emotions with positive valence during the conversations, since the lowest score (4.89), achieved by the angry VH, is very closed to the scale neutral point of 5. Nevertheless, we achieved all the expected differences, accomplishing the goal of eliciting varying emotions in terms of valence. To reduce the positive bias, stronger negative narratives may be designed, and the VH may be improved to increase the subject's empathy. In terms of arousal, the subjects did not report any differences. Therefore, we were not able to elicit coherent arousal reactions in subjects. One potential reason is that VHs did not change the prosody features of the voice depending on the emotional state such as volume, tone and speech rate, and it is well-known that para-verbal aspects of human communication carries on a significant portion of the emotional content expressed by individuals (Mehrabian & Ferris, 1967). This could affect the arousal perception as well as its elicitation. Further research is needed to elicit arousal during conversation with VHs and to explore its relationship with para-verbal features.

4.6. Emotion identification

In the case of emotional identification of the VHs, the expected differences were all achieved in valence. Therefore, the subjects were able to identify all the emotions designed in terms of valence. In the case of arousal, several differences are shown, supporting the hypothesis that although coherent arousal was not elicited in the subjects, they were able to identify different arousal patterns between VHs. In general, the negative emotions have been identified as higher arousal than the positive ones. It could be related to the negative bias in social-emotional interaction, since humans use negative information far more than positive information (Vaish, Grossmann, & Woodward, 2008). On the other hand, happy VH presented lower perceived arousal than expected as it was close to the scale neutral point, and further efforts are needed either to elicit effusive happy states, or to reduce the negativity bias effect.

4.7. Future of virtual humans

Recent reviews underscore the crucial role of ecological and social emotion elicitation methods in various fields, particularly healthcare, including the assessment, monitoring, and intervention of mental health disorders like depression, anxiety, bipolar, and psychotic disorders. Tudor Car, Dhinagaran, Kyaw, Kowatsch, Joty, Theng, and Atun (2020) highlighted the urgent need for comprehensive evaluation of diverse healthcare conversational agents, focusing on their acceptability, safety, and effectiveness. Their review revealed that most conversational agents described in the literature are text-based, machine learning-driven, and delivered via mobile apps. Torous et al. (2021) emphasized the significance of recent technological advancements in chatbots to mental health research and care. They detailed how smartphones, social media, artificial intelligence, and virtual reality offer new opportunities for "digital phenotyping" and remote intervention, but noted the need for further efforts towards solidifying implementation. Adikari et al. (2022) proposed an empathic conversational agent framework, combining natural language processing techniques and artificial intelligence algorithms for real-time monitoring and cofacilitation of patient-centered healthcare. However, their system relies on a rule-based emotional message generation, limiting the potential for open-ended conversation. Furthermore, they did not integrate the conversational agent with in an avatar. In a comprehensive review of emotional chatbots, May and Denecke (2020) combines VR, a chatbot and a voice user interface in a smartphone application, but without a Large Language Model. Bilquise, Ibrahim, Shaalan, et al. (2022) discussed the current limitations of the state-of-the-art. They pointed out that generative chatbots are still not based on LLM, possibly due to their novelty. Moreover, no research thus far has incorporated a conversational agent into an avatar. Denecke and May (2023) presented the most recent review of Conversational Agents in health, illustrating that the current landscape of health CAs is predominantly characterized by rule-based, simple systems in terms of CA personality and interaction. In other fields, Stanica, Dascalu, Bodea, and Moldoveanu (2018) developed a VR scenario with a chatbot to facilitate job interview training. Face recognition and sentiment analysis were utilized to analyze and provide feedback to the participant. Okonkwo and Ade-Ibijola (2021) provided a review of chatbots in education, outlining potential future areas of education that could benefit from this advanced AI technology. However, they have not yet presented any comprehensive VH.

Despite the rise of AI-based technologies in fields such as natural language processing and computer graphics, there remains a lack of studies in affective computing that analyze human emotional responses using these methods. There is no existing work that integrates advanced technologies like audio transcription, LLM, speech synthesis, lip synchronization, and VR to create a realistic VH capable of conducting real-time, open-ended conversations. The system developed in this study represents a significant advancement by merging these technologies. Recent reviews of chatbots, conversational agents, or VHs suggest they will have significant implications in several fields such as health and education. It is our hope that the present work can guide future applications of VH in these areas.

4.8. Limitations and future research

The present study has certain limitations that future research should aim to address. A significant constraint lies in the VR platform employed, which hampers the natural movements and gestures of the VH, diminishing its realism. The platform does not accommodate hand or leg movements, and the VH's blinking pattern was pre-programmed, limiting the organic behavior. Furthermore, the absence of an effective AI model to control voice modulation, rhythm, and tone could influence the participant's empathic response towards the VH. The static facial expression of the VH during the conversation also reduces the naturalism and realism of the interaction. Incorporating dynamic changes in the VH's facial state throughout the conversation could greatly enhance its lifelike quality. Another area of potential improvement pertains to the use of advanced language models, such as GPT-4, which could potentially provide superior text generation performance, thus improving the natural flow of conversation between the participant and the VH. Upgrades in audio transcription models could further enhance the realism of the VH's conversation. Looking forward, future research should concentrate on refining the VH to serve as a more ecologically valid social environment for eliciting psychological phenomena via emotions. Such efforts will aid in expanding the potential applications of VH across various disciplines. As we chart the course of our future research trajectory, we plan to expand our scope by developing emotion recognition systems grounded in biometric data already collected, including eye-tracking, EEG, ECG and EDA. Additionally, we aim to introduce a new phase in our experimental design by including subjects exhibiting depressive symptoms. Our objective is to architect an automated health assessment and monitoring system driven by supervised machine learning. This system will leverage our ecologically valid environment during emotional elicitation. Our hypothesis proposes that such systems will instigate the manifestation of specific phenomena tied to depressive symptoms. This is based on the understanding that depressive symptoms often correlate with impaired social skills and amplified negative emotional states, as supported by prior studies (Gross & Jazaieri, 2014; Segrin, 2000). Therefore, our approach aims to leverage these insights, advancing the development and application of intelligent systems for improved mental health diagnosis and treatment. In addition, further research needs to assess the influence of VH gender on emotion elicitation, as well as whether there are subject gender differences in terms of assessing the system.

5. Conclusion

The findings presented here mark a new step in the field of affective computing and its applications, presenting the first real-time conversational VH capable of engaging in semi-guided dialogue based on LLM. Drawing on a modular cognitive framework, we divided the VH into distinct mind and body components, providing a realistic life-size avatar with language comprehension, rational thought, creativity, and memory. We presented an extensive validation of the system in terms of technical performance, human–machine interaction, naturalness and realism, and emotion elicitation and identification. This breakthrough marks a considerable progression in the crafting of ecologically valid social virtual environments and the ability to generate emotion-driven responses that are shaped by the nuances of social and contextual factors. The intelligent system unveils a multitude of possibilities and potential applications, spanning mental health and permeating a plethora of other disciplines.

CRediT authorship contribution statement

Jose Llanes-Jurado: Conceptualization, Methodology, Investigation, Software, Formal analysis, Writing – original draft, Writing – review & editing. **Lucía Gómez-Zaragozá:** Conceptualization, Methodology, Investigation, Writing – review & editing. **Maria Eleonora Minissi:** Conceptualization, Methodology, Writing – review & editing. **Mariano Alcañiz:** Conceptualization, Resources, Writing – review & editing. **Javier Marín-Morales:** Conceptualization, Funding acquisition, Methodology, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

Thanks to Pepe Roda and Eduardo Manzano for modeling the avatars, and Luciano Spinella, Carmen Calero and Marina Catala for assisting the experiment.

Model availability

The conversational AI pipeline can be found in the following link https://github.com/ASAPLableni/LableniBOT.

Funding

This work was supported by the Vicerrectorado de Investigación de la Universitat Politècnica de València (UPV), Spain [Ayuda a Primeros Proyectos de Investigación (PAID-06-22)] and Ministerio de Derechos Sociales y Agenda 2030, Spain [Subvenciones para la realización de actividades relacionadas con la promoción e implementación de la Agenda 2030 para el Desarrollo Sostenible en España (A203023)].

Appendix A. Supplementary materials

This work provides an overview of the project and the experimentation in the following link https://youtu.be/Gu3KmhmF6WA.

Appendix B. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.eswa.2024.123261.

References

Abbaschian, B. J., Sierra-Sosa, D., & Elmaghraby, A. (2021). Deep learning techniques for speech emotion recognition, from databases to models. *Sensors*, *21*(4), 1249.

Abd-alrazaq, A. A., Alajlani, M., Alalwan, A. A., Bewick, B. M., Gardner, P., & Househ, M. (2019). An overview of the features of chatbots in mental health: A scoping review. *International Journal of Medical Informatics*, *132*, Article 103978. http://dx.doi.org/10.1016/j.ijmedinf.2019.103978, URL: https://www.sciencedirect.com/science/article/pii/S1386505619307166.

Adikari, A., De Silva, D., Moraliyage, H., Alahakoon, D., Wong, J., Gancarz, M., et al. (2022). Empathic conversational agents for real-time monitoring and co-facilitation of patient-centered healthcare. *Future Generation Computer Systems*, *126*, 318–329.

Angga, P. A., Fachri, W. E., Elevanita, A., Suryadi, & Agushinta, R. D. (2015). Design of chatbot with 3D avatar, voice interface, and facial expression. In *2015 international conference on science in information technology* (pp. 326–330). http://dx.doi.org/10.1109/ICSITech.2015.7407826.

Balano, J. B., Huerto, V. L., Sanchez, S., Saharkhiz, A., & De Goma, J. (2019). Determining the level of depression using BDI-II through voice recognition. In *2019 IEEE 6th international conference on industrial engineering and applications* (pp. 387–391). IEEE.

Bilquise, G., Ibrahim, S., Shaalan, K., et al. (2022). Emotionally intelligent chatbots: A systematic literature review. *Human Behavior and Emerging Technologies*, *2022*.

Borrego, A., Latorre, J., Alcañiz Raya, M., & Llorens, R. (2018). Comparison of oculus rift and HTC vive: Feasibility for virtual reality-based exploration, navigation, exergaming, and rehabilitation. *Games for Health Journal*, *7*, http://dx.doi.org/10.1089/g4h.2017.0114.

Borrego, A., Latorre, J., Llorens, R., Alcañiz, M., & Noé, E. (2016). Feasibility of a walking virtual reality system for rehabilitation: Objective and subjective parameters. *Journal of NeuroEngineering and Rehabilitation*, *13*, http://dx.doi.org/10.1186/s12984-016-0174-1, URL: https://jneuroengrehab.biomedcentral.com/articles/10.1186/s12984-016-0174-1.

Bradley, M. M., & Lang, P. J. (1994). Measuring emotion: The self-assessment manikin and the semantic differential. *Journal of Behavior Therapy and Experimental Psychiatry*, *25*(1), 49–59.

Bradley, M. M., & Lang, P. J. (2007). The International Affective Picture System (IAPS) in the study of emotion and attention.

Bredin, H., & Laurent, A. (2021). *End-to-end speaker segmentation for overlap-aware resegmentation*. arXiv, http://dx.doi.org/10.48550/ARXIV.2104.04045, URL: https://arxiv.org/abs/2104.04045.

Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., et al. (2020). Language models are few-shot learners. CoRR abs/2005, arXiv:2005.14165, URL: https://arxiv.org/abs/2005.14165.

Burden, D., & Savin-Baden, M. (2019). *Chapman & Hall / CRC artificial intelligence and robotics series*, *Virtual humans: today and tomorrow*. CRC Press, URL: https://books.google.es/books?id=Ue5jvAEACAAJ.

Ciechanowski, L., Przegalinska, A., Magnuski, M., & Gloor, P. (2019). In the shades of the uncanny valley: An experimental study of human–chatbot interaction. *Future Generation Computer Systems*, *92*, 539–548. http://dx.doi.org/10.1016/j.future.2018.01.055, URL: https://www.sciencedirect.com/science/article/pii/S0167739X17312268.

Clemente, M., Rodríguez, A., Rey, B., & Alcañiz, M. (2014). Assessment of the influence of navigation control and screen size on the sense of presence in virtual reality using EEG. *Expert Systems with Applications*, *41*, http://dx.doi.org/10.1016/j.eswa.2013.08.055, URL: https://www.sciencedirect.com/science/article/abs/pii/S0957417413006763?via%3Dihub.

Denecke, K., & May, R. (2023). Developing a technical-oriented taxonomy to define archetypes of conversational agents in health care: Literature review and cluster analysis. *Journal of Medical Internet Research*, *25*, Article e41583.

DeVault, D., Artstein, R., Benn, G., Dey, T., Fast, E., Gainer, A., et al. (2014). SimSensei kiosk: A virtual human interviewer for healthcare decision support. In *13th international conference on autonomous agents and multiagent systems, vol. 2* (pp. 1061–1068).

Diemer, J., Alpers, G. W., Peperkorn, H. M., Shiban, Y., & Mühlberger, A. (2015). The impact of perception and presence on emotional reactions: A review of research in virtual reality. *Frontiers in Psychology*, *6*, 26.

Ekman, P., & Friesen, W. V. (1978). Facial action coding system. *Environmental Psychology & Nonverbal Behavior*.

Galin, D., Ornstein, R., Herron, J., & Johnstone, J. (1982). Sex and handedness differences in EEG measures of hemispheric specialization. *Brain and Language*, *16*(1), 19–55.

Greene, S., Thapliyal, H., & Caban-Holt, A. (2016). A survey of affective computing for stress detection: Evaluating technologies in stress detection for better health. *IEEE Consumer Electronics Magazine*, *5*(4), 44–56. http://dx.doi.org/10.1109/MCE.2016.2590178.

Gross, J. J., & Jazaieri, H. (2014). Emotion, emotion regulation, and psychopathology: An affective science perspective. *Clinical Psychological Science*, *2*(4), 387–401.

Guillén-Riquelme, A., & Casal, G. (2011). Actualización psicométrica y funcionamiento diferencial de los ítems en el State Trait Anxiety Inventory (STAI). *Psicothema*, *23*.

Hachman, M. (2014). Battle of the digital assistants: Cortana, Siri, and Google Now. *PC World*, *32*(6), 13.

Hube, N., Vidackovic, K., & Sedlmair, M. (2022). Using expressive avatars to increase emotion recognition: A pilot study. In *CHI EA '22*, *Extended abstracts of the 2022 chi conference on human factors in computing systems*. New York, NY, USA: Association for Computing Machinery, http://dx.doi.org/10.1145/3491101.3519822.

Hutchens, J. L., & Alder, M. D. (1998). Introducing megahal. *CoNLL*.

Imam, I. F., & Kodratoff, Y. (1997). Intelligent adaptive agents: A highlight of the field and the AAAI-96 workshop. *AI Magazines*, *18*(3), 75–80. http://dx.doi.org/10.1609/aimag.v18i3.1307.

Koelstra, S., Muhl, C., Soleymani, M., Lee, J.-S., Yazdani, A., Ebrahimi, T., et al. (2012). DEAP: A database for emotion analysis ;using physiological signals. *IEEE Transactions on Affective Computing*, *3*(1), 18–31. http://dx.doi.org/10.1109/T-AFFC.2011.15.

Lang, P. (1980). Behavioral treatment and bio-behavioral assessment: Computer applications. In *Technology in mental health care delivery systems* (pp. 119–137). Ablex Publishing Corporation.

Ludlow, C. (2015). Central nervous system control of voice and swallowing. *Journal of clinical neurophysiology : official publication of the American Electroencephalographic Society*, *32*, 294–303. http://dx.doi.org/10.1097/WNP.0000000000000186.

Marín-Morales, J., Higuera Trujillo, J. L., Greco, A., Guixeres, J., Llinares, C., Scilingo, E., et al. (2018). Affective computing in virtual reality: emotion recognition from brain and heartbeat dynamics using wearable sensors. *Scientific Reports*, *8*, http://dx.doi.org/10.1038/s41598-018-32063-4.

May, R., & Denecke, K. (2020). Extending patient education with CLAIRE: An interactive virtual reality and voice user interface application. In *Addressing global challenges and quality education: 15th European conference on technology enhanced learning, EC-TEL 2020, Heidelberg, Germany, September 14–18, 2020, Proceedings 15* (pp. 482–486). Springer.

Mehrabian, A., & Ferris, S. R. (1967). Inference of attitudes from nonverbal communication in two channels. *Journal of consulting psychology*, *31*(3), 248.

Miao, F., Kozlenkova, I. V., Wang, H., Xie, T., & Palmatier, R. W. (2022). An emerging theory of avatar marketing. *Journal of Marketing*, *86*(1), 67–90.

Moemeka, E., & Moemeka, E. (2015). Leveraging cortana and speech. In *Real world windows 10 development* (pp. 471–520). Springer.

Nardelli, M., Valenza, G., Greco, A., Lanata, A., & Scilingo, E. P. (2015). Recognizing emotions induced by affective sounds through heart rate variability. *IEEE Transactions on Affective Computing*, *6*(4), 385–394. http://dx.doi.org/10.1109/TAFFC.2015.2432810.

Okonkwo, C. W., & Ade-Ibijola, A. (2021). Chatbots applications in education: A systematic review. *Computers and Education: Artificial Intelligence*, *2*, Article 100033.

Ospina-Bohórquez, A., Rodríguez-González, S., & Vergara-Rodríguez, D. (2022). A review on multi-agent systems and virtual reality. In *Distributed computing and artificial intelligence, volume 1: 18th international conference 18* (pp. 32–42). Springer.

Picard, R. W. (2003). Affective computing: Challenges. *International Journal of Human-Computer Studies*, *59*(1), 55–64. http://dx.doi.org/10.1016/S1071-5819(03)00052-1, URL: https://www.sciencedirect.com/science/article/pii/S1071581903000521, Applications of Affective Computing in Human-Computer Interaction.

Radford, A., Kim, J. W., Xu, T., Brockman, G., McLeavey, C., & Sutskever, I. (2022). Robust speech recognition via large-scale weak supervision. arXiv preprint arXiv:2212.04356.

Riva, G., Mantovani, F., Capideville, C. S., Preziosa, A., Morganti, F., Villani, D., et al. (2007). Affective interactions using virtual reality: The link between presence and emotions. *Cyberpsychology & Behavior*, *10*(1), 45–56.

Russell, J. A. (1980). A circumplex model of affect.. *Journal of Personality and Social Psychology*, *39*(6), 1161.

Safayari, A., & Bolhasani, H. (2021). Depression diagnosis by deep learning using EEG signals: A systematic review. http://dx.doi.org/10.20944/preprints202107.0028.v1.

Segrin, C. (2000). Social skills deficits associated with depression. *Clinical Psychology Review*, *20*(3), 379–403.

Serban, I. V., Klinger, T., Tesauro, G., Talamadupula, K., Zhou, B., Bengio, Y., et al. (2016). Multiresolution recurrent neural networks: An application to dialogue response generation. http://dx.doi.org/10.48550/ARXIV.1606.00776, URL: https://arxiv.org/abs/1606.00776.

Sollfrank, T., Kohnen, O., Hilfiker, P., Kegel, L. C., Jokeit, H., Brugger, P., et al. (2021). The effects of dynamic and static emotional facial expressions of humans and their avatars on the EEG: An ERP and ERD/ERS study. *Frontiers in Neuroscience*, *15*, http://dx.doi.org/10.3389/fnins.2021.651044, URL: https://www.frontiersin.org/articles/10.3389/fnins.2021.651044.

Stanica, I., Dascalu, M.-I., Bodea, C. N., & Moldoveanu, A. D. B. (2018). VR job interview simulator: Where virtual reality meets artificial intelligence for education. In *2018 zooming innovation in consumer technologies conference* (pp. 9–12). IEEE.

Suparatpinyo, S., & Soonthornphisaj, N. (2023). Smart voice recognition based on deep learning for depression diagnosis. *Artificial Life and Robotics*, 1–11.

Susindar, S., Sadeghi, M., Huntington, L., Singer, A., & Ferris, T. (2019). The feeling is real: Emotion elicitation in virtual reality. In *Proceedings of the human factors and ergonomics society annual meeting, vol. 63* (pp. 252–256). http://dx.doi.org/10.1177/1071181319631509.

Tack, A., & Piech, C. (2022). The AI teacher test: Measuring the pedagogical ability of blender and GPT-3 in educational dialogues. In A. Mitrovic, & N. Bosch (Eds.), *Proceedings of the 15th international conference on educational data mining* (pp. 522–529). Durham, United Kingdom: International Educational Data Mining Society, http://dx.doi.org/10.5281/zenodo.6853187.

Torous, J., Bucci, S., Bell, I. H., Kessing, L. V., Faurholt-Jepsen, M., Whelan, P., et al. (2021). The growing field of digital psychiatry: Current evidence and the future of apps, social media, chatbots, and virtual reality. *World Psychiatry*, *20*(3), 318–335.

Tudor Car, L., Dhinagaran, D. A., Kyaw, B. M., Kowatsch, T., Joty, S., Theng, Y.-L., et al. (2020). Conversational agents in health care: Scoping review and conceptual analysis. *Journal of Medical Internet Research*, *22*(8), Article e17158.

Vaish, A., Grossmann, T., & Woodward, A. (2008). Not all emotions are created equal: The negativity bias in social-emotional development. *Psychological Bulletin*, *134*(3), 383.

Weizenbaum, J. (1966). ELIZA—A computer program for the study of natural language communication between man and machine. *Communications of the ACM*, *9*(1), 36–45. http://dx.doi.org/10.1145/365153.365168.